

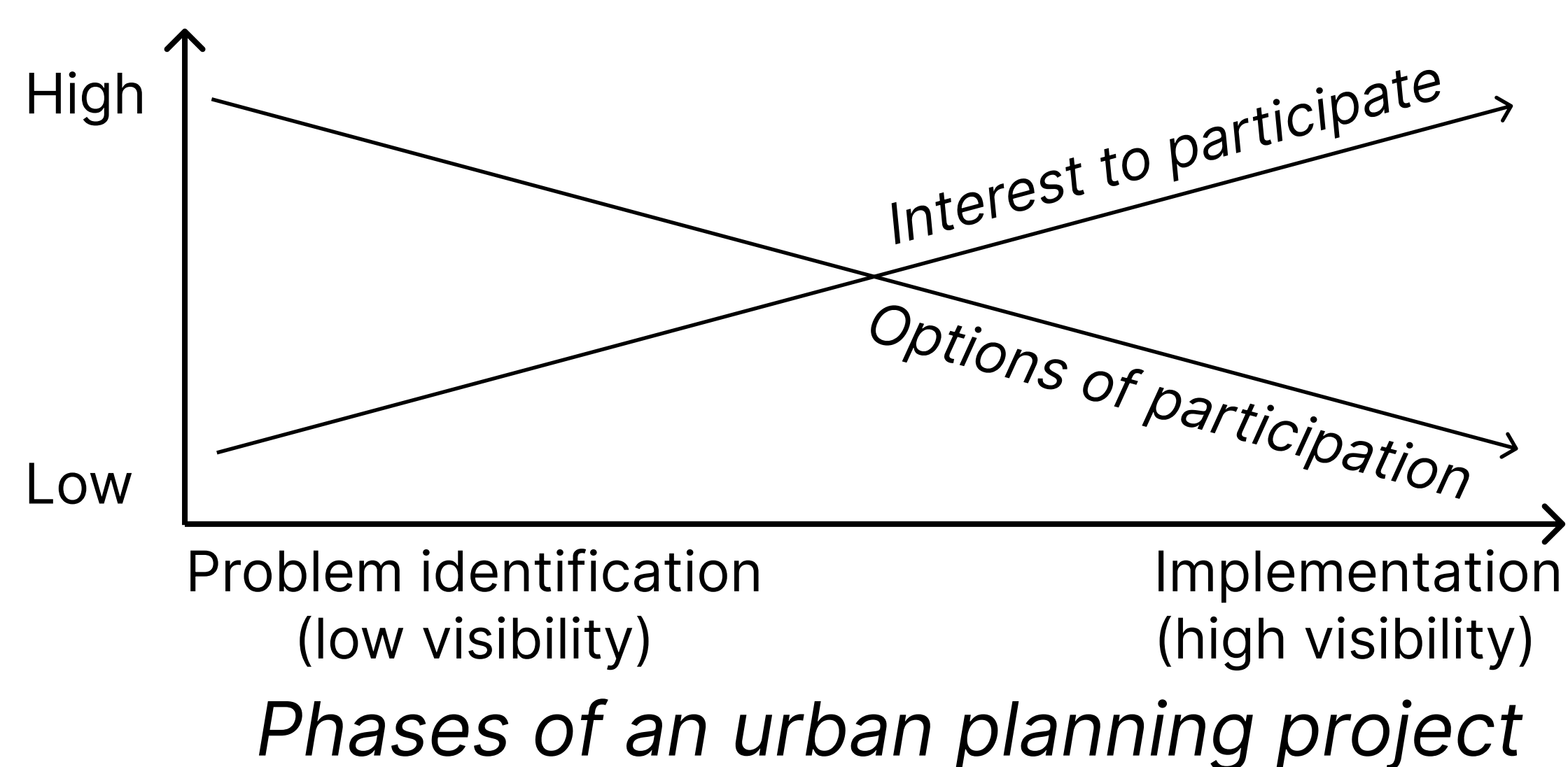
Augmented Reality-based Data Storytelling for Citizen Participation in Urban Planning

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Introduction

We explore applying augmented reality-based data storytelling to alleviate two difficulties:

Paradox of participation [1]



Rational ignorance [2]

Cost of learning related knowledge higher than the benefits they can gain

⇒ **Citizens give up** participating

Initial Design Requirements

Focusing on small-scale public space design, the design requirements are drawn from the literature in citizen participation, urban planning, and data storytelling:

Apply the martini glass narrative structure

Narratives should explain the context first, followed by interactive exploration of design alternatives.

Support interactive exploration of design alternatives

Visualizations and narratives should update dynamically to reflect the consequences of changing designs.

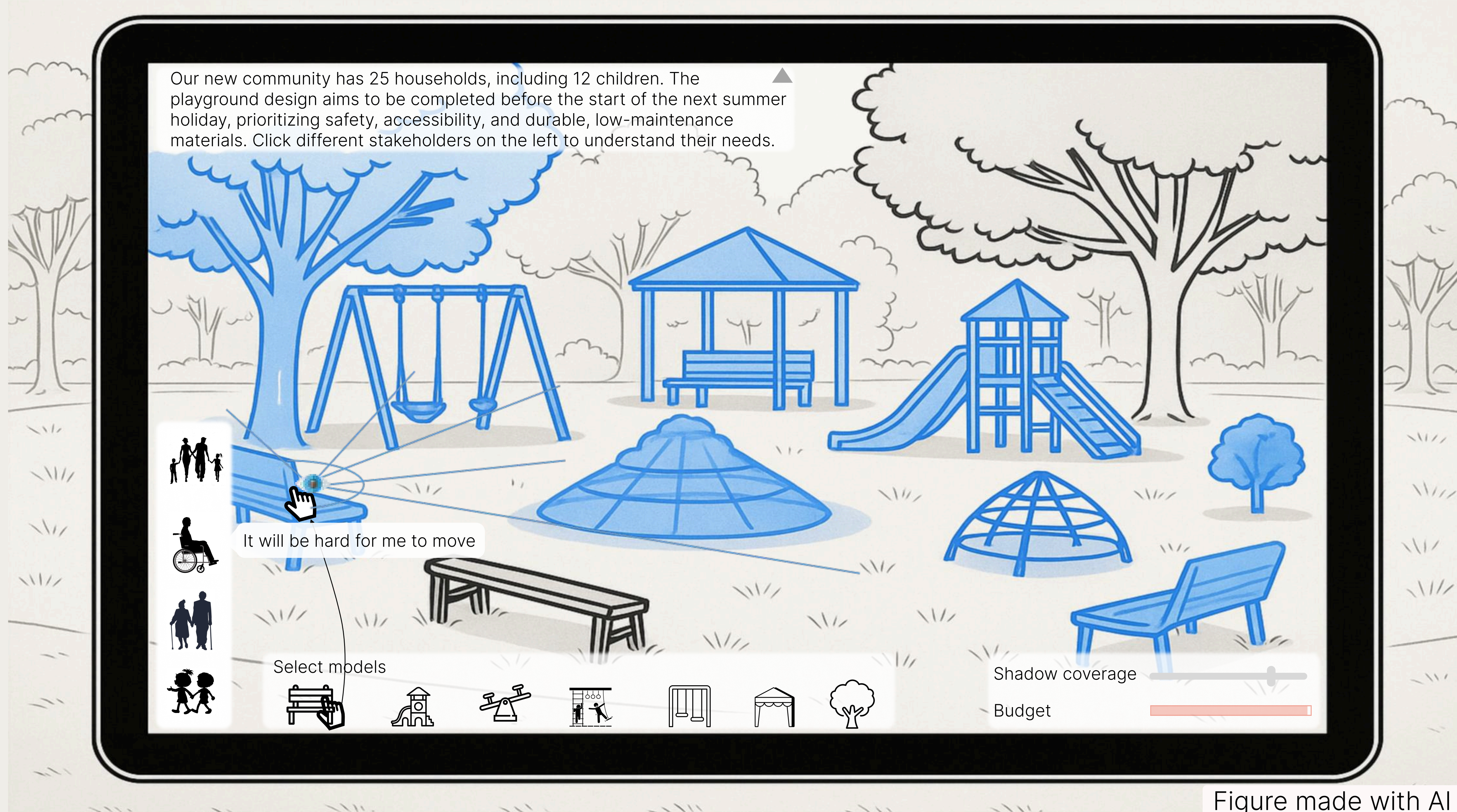


Figure made with AI

Link design choices to effects on lives

Narratives should explicitly explain how design alternatives affect real-life experiences.

Facilitate mutual understanding

Narratives should clarify how design alternatives impact relevant stakeholders.

Visualize people and activities concretely

Visualizations should represent how people would be affected by design alternatives in a concrete way.

Integrate abstract and concrete visualizations

Visualizations should integrate concrete visuals (e.g., 3D buildings) and abstract metrics (e.g., greening rate).

Next steps

Concretize and implement those design considerations in real-life applications.

Conduct evaluations to assess if the AR-based data storytelling approaches enhance citizen participation.

References

- [1] Wolf, Mario, Heinrich Söbke, and Florian Wehking. "Mixed reality media-enabled public participation in urban planning: A literature review." *Augmented reality and virtual reality: Changing realities in a dynamic world* (2020): 125-138.
- [2] Krek, Alenka. "Rational ignorance of the citizens in public participatory planning." 10th symposium on information-and communication technologies (ICT) in urban planning and spatial development and impacts of ICT on physical space, CORP. Vol. 5. 2005.